BOTSWANA DIGITAL PUBLIC PROCUREMENT TENDERING SYSTEM

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CHAPTER 1

INTRODUCTION

BOTSWANA DIGITAL PUBLIC PROCUREMENT TENDERING SYSTEM

Digitization of the Public Procurement Tendering process in Botswana through an online platform

Commented [WB1]: Give an introduction - defining your topic - what a procurement tendering system is - the background of the topic with reference to the current tendering system - introduce the topic. Not just a single line.

Commented [WB2R1]: Introduce what are tendering systems, why are they needed.

PROBLEM STATEMENT

Botswana public procurement process currently operates through a predominantly manual, paper-based process that follows the staged framework outlined in the PPRA Operation Manual (last issued Nov 5 2024) creating significant inefficiencies.

Commented [WB3]: Abbreviations used for the first time must be fully defined.

1. Time-Consuming Procedures: Each stage from procurement planning to contract management needs physical documentation, manual approvals and in-person submissions leading to prolonged tender cycles.
2. Limited Accessibility: Prospective bidders especially young entrepreneurs and small businesses face geographical and logistical barriers to accessing tendering opportunities, documents and submission channels.
3. Transparency gaps: The current system makes it difficult for the officers to track tender status, view evaluation criteria, understand award decisions or even access historical procurement data reducing public trust.
4. Administrative Workload: Procuring Entities struggle with manual record-keeping, compliance tracking, report generation and communication with bidders.
5. Inefficient Complaint Handling: The manual process for submitting and tracking complaints and appeals is slow and lacks visibility for affected parties.

Commented [WB4]: These point, though general for manual tendering systems, they are not connected to the problem. - you need to first give a background of how the current system works, then start to point where loopholes are coming from. Use paragraphs when explaining not point form - let it not look AI generated without your words explaining inline with the real situation on ground.

Commented [WB5R4]: Either you add another section titled "Background" then explain the current manual system so that when you write about the Problem statement you are pointing to real events within the system.

MOTIVATION

To digitize the procurement process and improve efficiency in the government workspace:

- Modernize Public Services: This project aligns directly with national development goals like Botswana's Vision 2036 puts emphasis on digital transformation and efficient governance.
- Promotes Inclusive Participation: A digital platform lowers barriers for young entrepreneurs and small businesses to participate in public tenders.
- Improves learning Opportunity: Developing this system provides practical experience in implementing real-world regulatory frameworks while solving a national challenge.
- Improves Accountability: The use of an online platform leaves audit trails which increases transparency in public spending, reducing potential for irregularities and building confidence.
- Creates Scalable Impact: A successful test could be expanded to become Botswana's e-procurement platform, serving all government entities.

Commented [WB6]: Write in paragraphs explaining in your words what these points mean connecting to the system you want to build.

PROPOSED SOLUTION

Developing a Digital Public Procurement Tendering Platform that mirrors the current country's procurement process while adding efficiency through automation and accessibility. The platform will feature:

1. A Public Bidder Portal for finding tenders, downloading documents, submitting bids online and tracking application status.
2. Procuring Entity Dashboard for managing the whole tender lifecycle digitally from publishing notices and receiving submissions to evaluating bids and managing contracts
3. Step-by-step guide for first bidders, tutorials on completing PPRA forms

Commented [WB7]: Explain - defending the importance of your system.

OBJECTIVES

The system must be able to:

- Successfully upload and download documents
- Pay for bid documents online
- Process bids against set criteria and regulations

Commented [WB8]: Rephrase e.g objective 1) Allow users to download and upload tender application documents.

Scope of the project:

CHAPTER 2

LITERATURE REVIEW

Introduction

This chapter presents a comprehensive review of literature related to digital public procurement systems, with a focus on e-procurement platforms and their implementation in various contexts. It examines global trends in e-procurement processes and explores specific implementations in African countries and comparable developing economies. The chapter concludes by identifying the gaps in existing systems and establishing the foundation for the proposed BW Procurement System.

Conceptual Framework of E-Procurement

E-procurement is also known as electronic procurement, which refers to the use of information and communication technologies to conduct procurement activities electronically or online. According to the World Bank (2020), e-procurement encompasses the complete

procurement process or cycle including: public advertisement of tender, distribution of bidding documents, submission of bids, bid opening, evaluation, contract award, contract management, order processing, payment and performance monitoring.

Croom and Brandon Jones (2017) define e-procurement as "the integration of digital technologies into the procurement process to automate and streamline purchasing activities, reduce transaction costs and improve transparency and accountability."

Types of E-Procurement Systems

Literature identifies several categories of e-procurement systems (Thai, 2017). The different types are as follows:

- E-Tendering- online publication, distribution and submission of tenders mostly common in government procurement
- E-Auctioning- online real time bidding for contracts mostly used for commodities and standard goods
- E-Catalogues- digital product listings with pre-negotiated prices used in framework agreements and routine purchases
- E-Invoicing- electronic billing and payment processing used in post award contract management

Global Trends in E-Procurement

The adoption of e-procurement systems has accelerated globally over the past years greatly driven by technological advancement and the recognition of procurement's strategic role in the public financial management.

African context

African countries have made significant efforts in e-procurements adoption, though implementation varies widely across the continent. Kenya's IFMIS has improved transparency and reduced processing times by approximately 40% though challenges persist with system downtimes and limited rural connectivity (Odhiambo and Njeru, 2021). Rwanda's UMUCYO platform launched in 2017, digitizes the whole procurement lifecycle and has reduced tender processing time by 60% while increasing bidder participation by 45% mostly among small businesses (Ndagijimana, 2022). South Africa's eTender serves as a centralized platform for publishing government tenders, providing public access to opportunities and supplier registration. The system primarily digitizes the publication phase and not the whole procurement process (Nkosi and Petrorius, 2021). Botswana currently operates through a manual PPRA framework that requires digitization to address existing inefficiencies.

Benefits of E-Procurement

E-procurements systems offer a number of benefits with the three main ones being:

1. Transparency- by making tender information publicly accessible, creating automatic audit trails and enabling real-time monitoring
2. Efficiency gains- through reduced paperwork, elimination of unnecessary physical and faster processing times
3. Promotes Inclusivity- by lowering the barriers for small businesses and young entrepreneurs who previously faced geographical and informational disadvantages.

Challenges to E-Procurement

The most common challenges include technical issues like poor internet connectivity and system integration difficulties (Mwangi and Kimani, 2021). The inertia to change from staff accustomed to manual processes and limited digital skills also hinder adoption (Hunja, 2019). Legal frameworks may not recognize electronic submissions and implementation costs can huge, though return on investment typically materialize within 3-5 years (World Bank, 2020).

Gaps in Existing Literature and Systems

Limited research addresses Botswana's specific procurement context, youth entrepreneurship barriers in public tenders or integrated learning resources for the first-time bidders. Regarding system gaps, current platforms provide minimal bidder guidance, rarely offer feedback on rejected bids, lack youth-specific features and offer limited visibility to tracking complaints. The proposed system addresses these through structured feedback, targeted youth onboarding and transparent status updates.

METHODOLOGY

This outlines the systematic approach for developing Digital Botswana Public Procurement Tendering System. The methodology ensure efficient development, adherence to requirements and alignment with project milestones.

Development Methodology

The Agile Scrum methodology will be used because:

- Iterative Development- project milestones require progressive delivery
- Stakeholder Involvement- regular engagement with procurements officers and bidders ensures real-world importance
- Adaptability- requirements may keep on changing or evolving as understanding of PPRA regulations deepens
- Continuous Feedback- Weekly supervisor demonstrations align well with Scrum sprint reviews

REQUIREMENTS ANALYSIS

Data collection will transpire through a thorough review of PPRA Operation Manual (2024) to understand workflows and compliance requirements. Discussions with the procurement officers and young entrepreneurs and direct observation of current procurement processes will suffice as other ways to collect data.

Functional Requirements

Bidder

- The system must allow the bidder to create an account or register
- The system must allow the bidder to login
- The system must allow the bidder to logout
- The system must allow the bidder to delete their account
- The system must allow the bidder to edit their account details
- The system must allow the bidder to view tender opportunities
- The system must allow the bidder to pay for tender application forms
- The system must allow the bidder to download tender documents
- The system must allow the bidder to submit bids/applications online
- The system must allow the bidder to track application status

Procurement Entity

- The system must allow the procuring entity to create an account
- The system must allow the procuring entity to login
- The system must allow the procuring entity to logout
- The system must allow the procuring entity to modify account details
- The system must allow the procuring entity to publish tenders
- The system must allow the procuring entity to manage tender evaluations
- The system must allow the procuring entity to view bidder submissions
- The system must allow the procuring entity to manage contracts

Administrator (PPRA)

- The system must allow the administrator to login
- The system must allow the administrator to logout
- The system must allow the administrator to manage users
- The system must allow the administrator to access all system aspects
- The system must allow the administrator to manage tenders and contracts
- The system must allow the administrator to monitor system activities
- The system must automatically process bids against tender criteria and regulations
- The system must shortlist top bids for procuring entity review
- The system must validate bid compliance with PPRA regulations

Non-functional Requirements

- The system must protect data from unauthorized access by using access restriction(Role-based access control, Attribute-based access control and Multi-Factor authentication)
- The system should respond quickly to user's request
- The system must handle a large number of concurrent users and active bids
- The system must be able to accommodate a growing number of users
- The system must be available 24/7 with minimal downtime
- The system must be safe and secure to store and protect data from unauthorized access
- The system must perform well with the payment gateway
- The system must be easy to maintain and update